

Tips For A Better Signal

1. Interference

2.4 GHz is the frequency used by most Wi-Fi devices. Unfortunately, this frequency is rather common: it's used by a few household devices, namely cordless phones and microwaves. Any Bluetooth device will also interfere with Wi-Fi signals, as will a cordless mouse.

2. Attenuation

Here are some common signal-scramblers. Every household will have most of these.

Plasterboard wall: 3dB

Glass wall with metal frame: 6dB

Wooden partition: 4dB

Office window: 3dB

Metal door: 6dB

Metal door in brick wall: 12.4dB

Now you know what happened to that signal! We're assuming average thickness here. The thicker you go, the lower your signal strength becomes. You cannot avoid attenuation from most of these materials, but knowing what material absorbs what may help you better position the router/AP.

3. Up, up and away

Heights are good, especially for wireless devices. Your router could even be mounted on top of a cupboard (just ensure it's away from walls if possible). Creating a little cradle mount dangling from your ceiling would be better. For a PCI wireless card, the metal cabinet and components will cause attenuation. No way to avoid this, but keeping your cabinet on a table (as opposed to on the floor) will help reception. PCMCIA cards in laptops will have the most problems with signals, and it's better to use the integrated Wi-Fi solution, as the antenna will probably extend into the screen area.

4. Higher gain does not necessarily mean a better signal

Higher gain means the signal is concentrated more in a particular direction. Note that the power supplied remains unchanged, just that the same amount of energy (beam) is focused in a much tighter area. Higher gain antennas therefore provide more

focused signal beams. They're better for large open areas, and may be less effective in case the site has a lot of attenuation-creating walls, furniture, etc.

5. Other attenuating culprits

Tinted windows will absorb a lot of your wireless signal. Note that regular glass windows will not. Even wire meshes and metal grilles will absorb signals like crazy. The degree of attenuation may be as much as what a solid metal object causes.

6. Polarity

By polarisation here, we mean the relation between the physical orientation of each antenna and the electric field that is parallel to the antenna. Omni-directional antennas have vertical polarisations due to the fact that they're meant to stand vertically. This is why most PCMCIA cards don't perform well in tandem with them (remember they slot into your laptop horizontally).

Senao ECB 3220

A Bridge for all occasions

Senao's ECB 3220 is positioned as a wireless bridge while also functioning as an AP. In fact, in keeping with its primary function (bridge), the ECB 3220 had the best signal in Zone 2, and just missed top spot in Zone 3. It's wall mountable.

Although Senao's firmware is reasonably laid out, it lacks detailed help on some settings. The basics are covered well though, but we had a few problems setting it up.

The ECB 3220s a steady performer. It didn't lose reception at all throughout the tests, even while moving between Zones—except while moving up the staircase to Zone 3.

You should know that all the devices in this test, with the exception of the ASUS WL-600G, lost reception while the laptop was being moved between zones. This tells us that the ECB 3220 has a very strong signal.

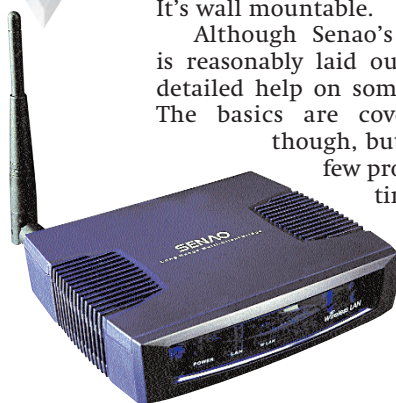
Carrying a price tag of Rs 8,280, the ECB 3220 is as expensive as an AP. However, because of its powerful signal, office users will find it useful as a bridge between two wireless networks. We're awarding it our Best Buy Silver award.

New Horizons

Throughout these tests, we hope one thing's become crystal clear... wireless devices, though offering freedom from wires, cannot sustain anywhere near the high data transmission rates that good old Ethernet can. However, Wi-Fi as a technology has matured sufficiently to allow us to surf the Net and stream audio and video sans the discomfort. And we're talking discomfort from iffy connections, sporadic transmission problems, loss of signal, etc. 54 Mbps may seem light years away from the actual transfer rates these devices offer when really stressed.

The silver lining on the wireless cloud promises to be the 802.11n standard. As of now, details about the exact specifications are sketchy (there's no formal draft as of today). What we do know is the intensity of signal will be much stronger than the current "g" standard. Also promised is nearly ten times the bandwidth—both the maximum theoretical and the typical (see table *Wi-Fi Standards*). If 802.11n realises this, we fully expect to see Wi-Fi penetration soar in our country as more and more people take the wireless plunge. ■

michael_browne@thinkdigit.com



Contact Sheet		Wireless Devices	
Brand	Company	Phone	E-mail
ASUS	ASUSTek Comp Inc.	022-40058923	mamta.bhatia@asus.com
Buffalo	Advent Infotech Pvt. Ltd	011-41436751	buffalo@adventelectronics.com
Compex	Rashi Peripherals India Pvt Ltd	022-67090909	navinderc@rptechindia.com
D-Link	D-Link India Ltd	022-26526696	sales@dlink.co.in
Level One	Minds India Pvt Ltd	011-26312095	foziayasin@gmail.com
Linksys	Ingram Micro India Pvt. Ltd	9323112279	sunil.z@ingrammicro.co.in
Netgear	Rashi Peripherals India Pvt Ltd	022-67090909	navinderc@rptechindia.com
Senao	Rashi Peripherals India Pvt Ltd	022-67090909	navinderc@rptechindia.com